

Weather Forecasting Satellites—Past, Present, & Future

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Abstract

Climate change has made weather more erratic and unpredictable. As a result, a growing need to develop more reliable short-term weather prediction models paved the way for a new era in satellite instrumentation technology, where radar systems for meteorological applications became critically important. This paper presents a comprehensive review of the evolution of weather forecasting satellites. We trace the technological development from the early weather and climate monitoring systems of the 1960s. Since the use of stabilized TV camera platforms on satellites aimed at capturing cloud cover data and storing it on magnetic tape for later readout and transmission back to ground stations, satellite sensor instrument technologies took great strides in the following decades, incorporating advancements in image and signal processing into satellite imagery methodologies. As innovative as they were, these technologies still lacked the capabilities needed to allow for practical use cases other than scientific research. The paper further examines how the next phase of satellite platforms is aimed at addressing this technological gap by leveraging the advantages of low Earth orbit (LEO) based satellite constellation deployments for near-real-time tracking of atmospheric hydrometers and precipitation profiles through innovative methods. These methods involve combining the collected data into big-data lakes on internet cloud platforms and constructing innovative AI-based multi-layered weather prediction models specifically tailored to remote sensing. Finally, we discuss how these recent advancements form the basis for new applications in aviation, severe weather readiness, energy, agriculture, and beyond.

Keywords: space-based remote sensing; radar; multi-sensor satellite constellations; neural networks; weather forecasting models; cloud technology; radiometers



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1. Introduction

In 2014, Neil deGrasse Tyson, a well-renowned Astrophysicist, defined the weather phenomena as what the atmosphere does in the short term, on an hourly or daily basis [1]. This description draws support from the butterfly effect, a concept derived from the chaos theory, which states that a system's sensitive dependence on initial conditions can cause small changes in one possible state—within a given state-space defining a deterministic nonlinear system—to produce large differences in a future state, conditioned by some probability of occurrence [2]. Sensitive dependence, as Edward Lorenz—one of the founding fathers of chaos theory—has stated [3], is the chief cause of our well-known failure to make nearly perfect weather forecasts [4]. That is why weather behavior is harder to predict due to its chaotic nature, especially in comparison to climate, which is defined as the

long-term average of weather over a number of years, a long-term trend. Prior to the satellite era, the invention of radio transmitters and the radiosonde, a small box equipped with weather instruments and a radio transmitter, enabled improved monitoring capabilities for land-based weather observation stations worldwide by making them airborne and launching them on balloons to high altitudes—several miles above the ground [5]. A method that mildly improved weather forecasting observations [6]. At the start of the 19th century, a growing inference began to emerge as data accumulated from various environmental monitoring sources. This suggested that the initial assumption—that we need to defend ourselves from the environment, which led to the development of numerous monitoring capabilities—was only half the story, as evidence showed [7] that humankind is capable of unintentionally altering the environment in unfavorable ways [8].

Driven by the broad assumption that humanity needs to protect itself from the environment, coupled with the emerging ability to launch Earth-orbiting satellites equipped with advanced instruments capable of monitoring the weather on a global scale—President Kennedy urged leaders during the 1960s to support the establishment of the world weather watch (WWW). This initiative aimed to improve weather predictions, enhance event warning capabilities, and more. Only in the 1970s did the realization that humankind is able to influence the environment transition into policy. Under the directive of President Nixon, measures were taken with the aim of reversing the negative environmental trends and protecting the environment from further harm caused by human activities. One of the main triggers for the historical events outlined above was the launch of one of the first dedicated weather satellites, TIROS-1, equipped with an infrared observation system, on 1 April 1960, by the United States. Its objective was to test experimental sensory television techniques, utilizing cameras optimized for photographing the earth's cloud cover rather than land areas, as a means of presenting a very large amount of weather data from over a large part of the earth in a very short time frame. Post-launch, an initial performance evaluation concluded that the picture quality was excellent—exceeding those obtained during evaluation tests [9]—and demonstrated that the design of the onboard systems was extremely valuable as a meteorological research tool [10]. The value of satellite technology in meteorology stems from the inherent challenges of acquiring weather data from remote regions such as oceans and the Arctic. Prior to satellites, these data gaps severely limited the development of accurate weather forecasting models. Satellite systems have since enabled near-real-time, global atmospheric observations, significantly enhancing model reliability and coverage. Satellites provide a complementary data collection platform, offering enhanced coverage and capabilities that other tools—such as weather balloons, monitoring aircraft, and ships—cannot fully deliver [11].

In this paper, we present a review of recent advancements in satellite instrumentation technology, remote sensing applications, and their role in developing near-real-time weather prediction models for a range of civilian and military sectors.

2. Orbit Classification

Prior to discussing weather satellites, it is essential to understand their operational altitudes and orbital classifications. For this purpose, the following five types of planetary orbits are reviewed:

- (a) High Altitude Platforms (HAPs), including the stratospheric layer (~15–50 km altitude)
- (b) Low Earth Orbit (LEO)—approximately 300 to 2000 km altitude
- (c) Medium Earth Orbit (MEO)—approximately 4000 to 8000 km altitude
- (d) Geostationary Earth Orbit (GEO)—orbiting along the Earth's equatorial plane at ~35,786 km altitude

- (e) Highly Elliptical Orbit (HEO)—elliptical orbits with variable perigee and apogee, often used for extended regional coverage.

Placing weather monitoring instruments in the stratosphere, approximately 15 to 50 km above ground level, offers several advantages. Since most weather phenomena occur in the lower troposphere (0–12 km), positioning instruments on a high-altitude platform (HAP) in the stratosphere allows for effective monitoring of local weather conditions over the surface area below. HAPs operating at altitudes between 17 and 24 km, where wind velocities are minimal, can maintain a stable position using Global Positioning System (GPS) technology combined with a 360° low-thrust propulsion system onboard. This setup leverages the low-pressure conditions of the stratosphere. Additionally, HAPs provide near-zero latency in radio transmissions between the platform and ground stations [12,13].

Other weather monitoring instruments, such as satellites classified as operating at LEO, maintain the lowest possible delay in a radio transmission path between a satellite and any ground station. These satellites operate at altitudes ranging from 300 km to 2000 km, positioned below the first Van Allen radiation belt. Medium Earth Orbit (MEO) satellites function at altitudes between 4000 km and 8000 km and inherently experience longer signal delay times compared to Low Earth Orbit (LEO) satellites. Geostationary Earth Orbit (GEO) satellites have a zero-inclination angle, meaning their angular velocity matches the rotation rate of the Earth they orbit. This synchronization causes the satellite to appear stationary from the ground—hence the term ‘Geostationary’. This orbit is referred to as the equatorial plane or Clarke orbit, but it does not cover the polar regions of the Earth. The perigee and apogee of satellites in such orbits vary depending on the planet being orbited. The main advantage of maintaining this orbit is the larger area coverage due to its higher altitude, while the obvious drawback is the increased communication delay times. Highly Elliptical Orbit (HEO) satellites revolve around the planet located at one of the elliptical orbit’s focal points [14]. Satellite constellations are common at the LEO or MEO level, as the area coverage provided by a single satellite is pretty small due to its low altitude. For this reason, a group of satellites is organized to maintain inter-connections along with connections with ground stations or user agents, to achieve larger area coverage. In contrast, a GEO use case is simpler by comparison; one satellite is enough to achieve large surface coverage of the Earth, up to 42% of the planet’s surface, except the polar regions.

3. Satellite Sensor Instrument Technologies

This historical overview provides the foundation for a detailed examination of satellite sensor technologies, which have continuously advanced atmospheric observation capabilities. These technologies are discussed in Section 3.

The Automatic Picture Transmission (APT) TV Camera System for Meteorological Satellites [15] was installed on NIMBUS-1, a meteorological LEO satellite [16] launched on 28 August 1964. It was specifically designed to provide global coverage of Earth’s cloud formations and to support various aspects of atmospheric research. The satellite (see Figure 1) carried a three-camera system that operated during daylight hours, complemented by an infrared scanner for nighttime observations. The collected data, including cloud formation imagery, was stored on magnetic tape and transmitted continuously in near-real time via the onboard communication system. The cloud formation imaging process operates as follows.

Optical instruments onboard the satellite were utilized to focus on a specific cloud formation area directly beneath the satellite, projecting the image onto the face of a storage vidicon. The captured optical image was then converted into an electrical signal through an electron beam readout process (see Figure 2). This electrical signal was subsequently modulated onto a subcarrier, which in turn modulated the Very High Frequency (VHF)

transmitter. The signal was transmitted via the satellite's antenna to ground-based receiving stations, where the image was reconstructed in real time using a facsimile reproduction system. When positioned in a circular orbit at an altitude of 900 km, the APT camera's lens produced an image width of approximately 1720 km per side. Higher orbital altitudes allowed for larger surface area coverage within a single frame [15].

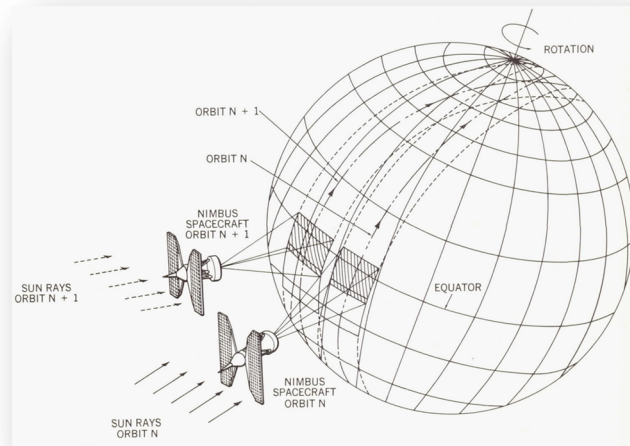


Figure 1. APT picture coverage—See Ref. [15].

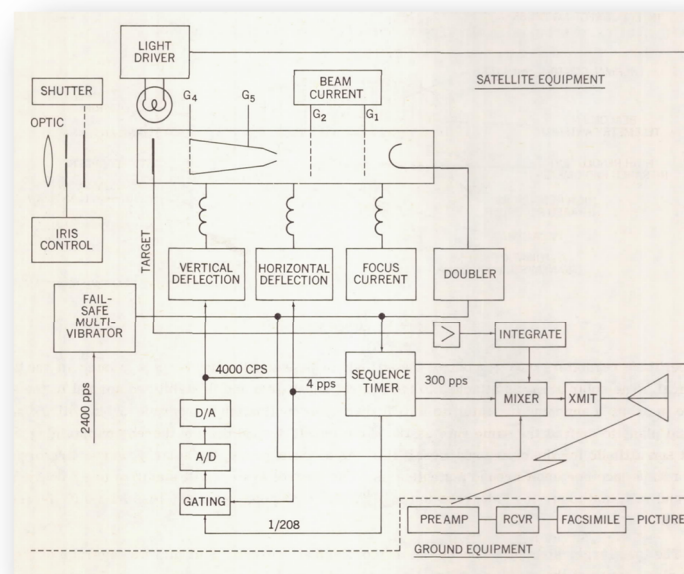


Figure 2. Block diagram of the APT system—See Ref. [15].

Microwave (MW) Remote Sensors

Due to a development trend that began as early as the 1960s with the launch of NIMBUS-1 and continued through subsequent missions such as the Landsat program in the 1970s, significant advancements were made in the fields of image acquisition and signal processing for satellite-based Earth observation. These advancements led to the emergence of a new class of sensors, broadly categorized into active and passive instruments. Compared to the early APT (Automatic Picture Transmission) systems of the 1960s, these modern sensors offer substantially improved spatial, spectral, and temporal resolution.

Each type operates based on distinct physical principles and serves specific applications in environmental monitoring, meteorology, and Earth science [17].

Radiometers are passive microwave sensors designed to measure the thermal emission of microwave signals originating from the sun and interacting with the Earth's surface. These signals are partially backscattered by the surface, with their characteristics influenced by the physical temperature, electrical properties, and composition of the target area or volume. Upon reception, this information is used to generate an emission profile of the target. Due to their reliance on solar illumination as the source of microwave energy, these instruments are often referred to as daylight sensors. Radiometers typically operate over a wide bandwidth and employ real-aperture antennas, which have large physical dimensions and are opaque to radiation at specific wavelengths [18]. Sounders, a derivative instrument of the radiometers sub-class, are specialized instruments designed for extracting vertical profiles of atmospheric parameters.

Active microwave sensors, such as radars, do not depend on daylight because they actively transmit microwave signals, eliminating reliance on the sun as a source. A meteorological radar is a typical example that uses real-aperture antennas to transmit modulated pulses. Through Doppler and range processing, it constructs backscatter images to detect rain, wind, snow, hail, clouds, ocean surface wave height, frequency and direction, temperature, wind speed, atmospheric moisture content, and vertical precipitation profiles [19]. This data is used for event warnings and short-range weather forecasts.

4. Understanding the Physical Principles of a Meteorological System

The principles underlying the design and operation of meteorological radar are essential for effective weather forecasting [20]. In this section, we introduce some general concepts related to this subject.

It begins with defining the electromagnetic interaction of EM waves with individual molecules. The total internal energy of a molecule can be categorized into three types of energy states—electronic, rotational, and vibrational energy:

$$\varepsilon_{\text{molecule energy}} = \varepsilon_e + \varepsilon_v + \varepsilon_r \quad (1)$$

These energy states define quantum states and, as a consequence, dictate how an atom responds to electromagnetic (EM) waves with energy values corresponding to the differences between energy levels, ΔE . This interaction leads to the creation of absorption and emission spectrum due to transitions between these energy states. Each atom has unique energy differences between states; therefore, when an EM wave with energy at specific frequencies interacts with an atom, absorption and scattering spectral lines are produced. From these spectra, one can infer which types of atoms constitute the molecules in question.

From a spectral graph, an absorption coefficient can be derived. Since spectral lines have tails that extend over a certain bandwidth, it is necessary to sum the contributions of all frequency lines. Based on this principle, a microwave propagation model was developed in the 1980s. This model computes the gaseous absorption coefficient, (K_g) as the sum of the oxygen absorption coefficient and the water vapor absorption coefficient for a clear-sky atmosphere.

That is,

$$k_g = k_{O_2} + k_{H_2O} \left[\frac{\text{dB}}{\text{km}} \right] \quad (2)$$

where $k_{O_2} = 0.182f \sum_{j=1}^{44} S_i F_i \left[\frac{\text{dB}}{\text{km}} \right]$, and $k_{H_2O} = 0.182f \sum_{j=1}^{34} (S_i F_i + S_c F_c) \left[\frac{\text{dB}}{\text{km}} \right]$.

The summation is performed over 44 oxygen spectral lines and 34 water vapor spectral lines, calculated as a function of line strength, S_i , and shape function, F_i . Both parameters are influenced by atmospheric conditions, specifically temperature (T), barometric pressure (P), and the partial pressure of water vapor (P_{H_2O}) at different altitudes (Z). In light of these principles, it becomes evident that the choice of operating frequency is critically important and depends on the specific atmospheric parameters one seeks to measure remotely using spectral absorption or emission lines.

While the discussion thus far has focused on clear-sky atmospheric conditions, the presence of extinction hydrometeors—such as rain, clouds, snow, or fog particles—introduces both absorption and scattering effects. In such cases, estimating the volume extinction coefficient becomes essential. This coefficient depends on the density, shape, size distribution, and dielectric properties of the particles within the volume interacting with the propagating electromagnetic (EM) waves.

To model this interaction, it is commonly assumed that the EM waves interact with spherical particles within the volume. This simplification allows for the analysis of the power density, S_i , of an incident EM wave upon a particle with cross-sectional area A and radius r (see Figure 3).

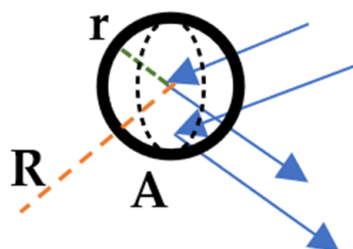


Figure 3. Self. Particles hit by EM waves. Source: Self.

A portion of the energy is absorbed, while the remainder is scattered in all directions within a radius R from the center of the particle outward (see Figure 3). As a consequence, we can define an absorption cross-section

$$Q_a = \frac{P_a}{S_i} \left[\text{m}^2 \right] \quad (3)$$

where P_a is the absorbed power. The power scattered by the particle at a distance R is calculated as the surface integral over a sphere with radius, R

$$P_s = \iint S_s(\theta, \phi) R^2 d\Omega \quad (4)$$

where $S_s(\theta, \phi)$ is the power density of radiation is scattered in directions (θ, ϕ) at a distance R from the particle. The scattering cross section is therefore defined as

$$Q_s = \frac{P_s}{S_i} \left[\text{m}^2 \right] \quad (5)$$

The total power extracted from the EM wave is $P_a + P_s$ and the extinction cross section is defined as

$$Q_e = Q_a + Q_s \quad (6)$$

Since we are interested in meteorological applications, our focus is on the backscattering power towards the radar— $S_b = S_s|_{\theta=\pi}$, acting as an active MW sensor. Therefore, the backscattering cross section can be defined as

$$\sigma_b = 4\pi R^2 \frac{\mathcal{S}_b}{\mathcal{S}_i} [\text{m}^2] \quad (7)$$

At this point, the discussion concerning a single particle can be extended to a volume containing many particles. In this case, the cross section of the entire volume is defined as the algebraic sum of the individual cross sections of all particles within that volume. Accordingly:

- (1) The volume scattering coefficient is defined as:

$$K_s = \int_{r_1}^{r_2} \widetilde{p(r)} Q_s(r) dr \quad (8)$$

- (2) The volume extinction coefficient for clouds is defined as:

$$K_{e_{\text{Clouds}}} = \int_{r_1}^{r_2} \widetilde{p(r)} Q_e(r) dr. \quad (9)$$

- (3) The volume backscattering coefficient (also called radar reflectivity) is defined as:

$$\sigma_V = \int_{r_1}^{r_2} \widetilde{p(r)} \sigma_b(r) dr \quad (10)$$

for a drop size distribution,

$$\widetilde{p(r)} = p(r) * [U(r_2) - U(r_1)] \quad r_1 \leq r \leq r_2 \quad (11)$$

the partial concentration of particles per unit volume and increment of variable radius r .

- (4) The calculation of the volume extinction coefficient (K_e) in the presence of precipitation (such as rain) follows a slightly different approach. It requires considering a specific drop-size distribution that characterizes the number of drops with radius r per unit volume. A well-established model for this purpose was developed by Marshall and Palmer in 1948. Their model, designed for rainfall intensities ranging from 1 to 23 mm/h at the ground surface, provides highly accurate results for estimating rain-induced extinction:

$$p_{\text{Rain}}(r) = N_0 e^{-2br} \quad (12)$$

where $N_0 = 8 * 10^6 \left[\frac{1}{\text{m}^4} \right]$, $b = 4100 * R_r^{-0.21}$ and R_r – rainfall rate $\left[\frac{\text{mm}}{\text{hr}} \right]$.

General Note: The calculation of these coefficients is, in reality, more complex than discussed thus far. Additional factors must be considered, including the wavelength (λ_0) dependence arising from Rayleigh or Mie scattering approximations. These approximations describe how electromagnetic waves are scattered by particles: Rayleigh scattering applies when the particles are much smaller than the wavelength, leading to a strong λ^{-4} dependence, while Mie scattering applies when the particles are comparable to or larger than the wavelength, (e.g., dust, cloud droplets), and exhibits a weaker and more complex wavelength dependence. Other influencing factors include the altitude (Z) of the particles within the atmosphere, the optical penetration depth, and various thermodynamic or compositional parameters. The altitude (Z) of the particles within the atmosphere, penetration depth, and other parameters.

The theory outlined in this section provides the foundation for a simplified form of the radar equation used in meteorological applications [21]. This equation offers a theoretical description of the principle underlying short-pulse radar operation—namely, the reception of power by the radar, which depends on the degree of backscattering from a collection of particles dispersed within a finite volume of the atmosphere, located at a distance R from

the radar. The received power is also influenced by the radar's antenna gain (G_0) and the transmitted wavelength (λ_0):

$$P_{received\ power} = \frac{P_t G_0^2 \lambda_0^2}{(4\pi)^3 R^4} \sigma_V e^{-2\tau} \quad (13)$$

where τ is the total one-way path attenuation between the radar and the volume at a distance (R):

$$\tau = \int_0^R (K_g + K_{e_{clouds}} + K_{e_{rain}}) dr \quad (14)$$

Neglecting factors such as antenna beamwidths at half-power, pulse duration, and other secondary parameters, the amount of received power corresponding to the transmitted frequencies toward a given volume provides an indication of the composition of that volume. This information contributes to constructing a precipitation profile (as in the rainfall example discussed) for a specific region of the atmosphere at a given moment in time. Thus, the theoretical description of the fundamental principles governing the operation of meteorological radars is concluded.

5. Weather Forecasting Models

We have discussed a variety of instruments and their data products, such as sounders and radars. However, effectively utilizing these instruments has posed a significant challenge due to one primary limiting factor that has yet to be adequately addressed. Referring back to our initial distinction between climate and weather, it becomes evident that the continuous acquisition of observational data is critical for developing short-term forecasting models for weather applications. In this context, the term satellite revisit rate describes the temporal frequency at which a satellite observes the same location on Earth. In other words, it represents the time interval between consecutive satellite observations of a specific geographic point. A higher revisit rate translates to an increased data refresh rate, thereby improving the accuracy, resolution, and practicality of near-real-time weather forecasting models and enabling new operational capabilities.

As illustrated in Table 1, to date, satellite revisit rates have been relatively low, limiting the effectiveness of such forecasting models.

Table 1. Historical Satellite Revisit Rates.

Satellite System	Revisit Rate	Date of Operation
TRMM	3 days	1997–2015
CloudSat	16 days	2006–2024 (decommissioned)
GPM	3 days	2014–2028
RainCube	Limited	2018–2021
EarthCare	16 days	2023
INCUS	Limited	Start—2027

Internalizing the critical role of revisit rates in the scalability and accuracy of forecasting models, and building upon decades of advancements in remote sensing, new satellite missions have been launched to deepen understanding of Earth's global water cycle and to investigate atmospheric convective systems, cyclic storms, and precipitation processes using radar and radiometer technologies.

One example is the launch of the Tropical Rainfall Measuring Mission (TRMM) to Low Earth Orbit (LEO) on 27 November 1997, a joint venture between NASA (Washington, DC,

USA) and the Japan Aerospace Exploration Agency (JAXA) (Tokyo, Japan). TRMM aimed to improve understanding of global energy and water cycles by observing the spatial and temporal distribution of tropical rainfall, along with associated hydrometeor structures and heating patterns over tropical and subtropical regions. This was accomplished using onboard instruments such as the precipitation radar (PR), which provided 3D imaging of rainfall structures and vertical distribution, as well as quantitative rainfall measurements over land and ocean.

The PR featured a 128-element active phased-array system operating at 13.8 GHz. The system's assembly included a transceiver with 128 solid-state power amplifiers and low-noise amplifiers (LNAs), which were selected at each time step by a circulator that periodically switched between transmission and reception while sharing the same antenna. This was followed by PIN-diode phase shifter elements integrated with dividers/combiners, a 16-branch divider/combiner, an intermediate driver amplifier, and a signal processing subsystem comprising a frequency converter, IF unit, and system control and data processing unit [22].

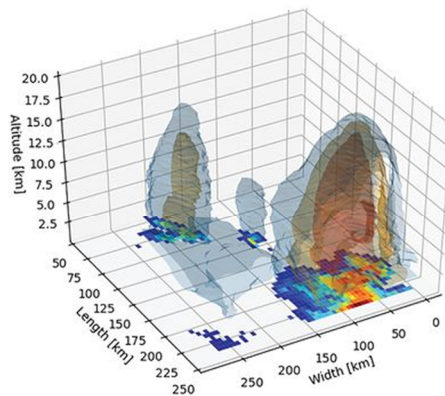
The NASA GOES-16 satellite, shown in Figure 4, is the first of the GOES-R series of Geostationary Operational Environmental Satellites (GOES), operated also by the National Oceanic and Atmospheric Administration (NOAA). Its Advanced Baseline Imager (ABI) consists of 16 spectral bands at visible and infrared wavelengths, providing high-resolution Geostationary Lightning Mapping (GLM) of the Earth [23]. Its payload consists also Earth and Sun weather monitoring instrumentations.



Figure 4. U.S Weather Forecasting Satellite (Wikipedia)—GOES 16 (formerly known as GOES-R).

A second instrument, the Microwave Imager (TMI), is a 9-channel radiometer operating at dual-polarized frequencies ranging from 10.7 GHz to 85.5 GHz, with an $11 \text{ km} \times 8 \text{ km}$ field of view at 37 GHz and high scanning capability. It complemented the precipitation radar by providing information on total hydrometeor content within precipitation systems (see Figure 5). The combination of active (PR) and passive (TMI) instruments, along with the inclined orbit of the TRMM satellite sampling the atmosphere throughout its orbit, enabled the generation of high-quality precipitation data relative to the operational capabilities available during TRMM's mission. This integration led to significant scientific advances by supplying additional input parameters to hydrological and land-surface models, enhancing our understanding of land-atmosphere interactions over timescales ranging from days to years [24].

3D Precipitation Structure



Example: 3D structure of the “champion storm”
from TRMM mission

Figure 5. 3D Precipitation Structure—Courtesy of Tomorrow.io.

Building on recent technological advances and leveraging the success of the TRMM mission, a next-generation satellite program was launched on 27 February 2014, as part of a continued partnership between NASA and JAXA. The Global Precipitation Measurement (GPM) core observatory aims to extend precipitation measurements to lower intensities (on the order of millimeters per hour) while expanding coverage beyond tropical and subtropical regions. By integrating a constellation of satellites operating in LEO, the GPM program achieves near-global precipitation observations approximately every three hours.

Compared to the instruments aboard TRMM, the GPM payload offers significantly enhanced precipitation measurement capabilities, providing detailed insights into the internal structure of cloud volumes, including hydrometeors and other atmospheric features. This is accomplished through the GPM’s Dual-Frequency Precipitation Radar (DPR) and GPM Microwave Imager (GMI).

The DPR’s Ku-band radar, operating at 13.6 GHz, is comparable to the TRMM’s Precipitation Radar (PR), but it introduces a second operating frequency at 35.5 GHz in the Ka-band. This higher frequency enables the detection of frozen precipitation and light rain, facilitating the generation of more detailed drop size distributions and improving the accuracy of rainfall and snowfall prediction models.

The GMI, a 13-channel scanning microwave imager, is designed to detect the backscattered power of electromagnetic waves (as described in earlier sections) and generate a layered representation of a target atmospheric volume. This capability enables the estimation of light rain and snowfall intensity by operating across a frequency range of 10 to 89 GHz, which is also used for the detection of moderate and heavy rainfall, similar to the functionality of the TRMM’s Precipitation Radar (PR). The added value of the GMI lies in its use of four additional high-frequency channels, ranging from 166 to 183 GHz, specifically intended for the measurement of moderate and light precipitation. This enhancement significantly improves the accuracy and resolution of precipitation forecasting models [25].

5.1. The Next Leap

The World Meteorological Organization (WMO) is a specialized agency of the United Nations (UN) tasked with establishing and maintaining an integrated Earth system observation network. This global network is composed primarily of state-operated weather and climate monitoring resources from the WMO’s 187 member states. Its objective is to provide reliable weather and climate data to a wide range of end-users [26]. For the first time, a

privately held entity will possess weather monitoring resources at a scale comparable to those of state-level infrastructure, contributing additional value to the WMO's operational capabilities and global observation efforts.

Tomorrow.io, a U.S.-Israeli startup specializing in advanced weather technology, is poised to revolutionize global weather and climate monitoring by deploying the world's first dedicated constellation of 30 weather observation satellites. The company aims to launch at least 20 of these satellites into low Earth orbit (LEO) by the end of 2025. The constellation will include two primary sensor types:

(a.) Active Ka-Band Radars

These radars will be placed at an altitude of 550 km in sub-synchronous, mid-inclination orbits. They will operate in the Ka-Band frequency range of 35.5 to 36 GHz, offering distinct advantages over lower-frequency systems. Specifically, Ka-Band operation enhances sensitivity to both low and high-intensity precipitation, improves detection of vertically integrated liquid water content compared to Ku-Band radars, and enables accurate estimation of a wide range of surface precipitation intensities. The shorter wavelength also allows for the use of smaller antenna and radio frequency (RF) components, optimizing satellite design. The radars will provide horizontal resolution of $5 \text{ km} \times 5 \text{ km}$ and vertical resolution of 250 m. Each radar is designed for a mission lifespan of approximately 5 years.

(b.) Passive Microwave Radiometers

These instruments will also be deployed at 550 km in mid-inclination orbits, operating within the 91 to 204 GHz frequency range. The radiometers will offer horizontal resolutions of $26 \text{ km} \times 28 \text{ km}$ at 91 GHz and $14 \text{ km} \times 17 \text{ km}$ at 204 GHz. The higher operating frequencies enhance the ability to detect atmospheric water vapor, cloud liquid water, and precipitation profiles. Each radiometer is designed with a mission lifespan of approximately 3 years.

This constellation will establish the first global, near-real-time sensing network capable of providing high-resolution precipitation measurements and atmospheric profiles. The resulting data will support a wide range of applications, including advanced severe weather alerts, early warning systems, and emergency management operations (as illustrated in Figure 6). For discussion purposes, the U.S National Weather Service defines tornadoes, severe thunderstorms, flash floods, and urban and small stream flooding (which is not life-threatening) as severe weather events. The service monitors these phenomena and issues warnings to the public, primarily to prevent any unfortunate loss of life [27].

For instance, it is well established that environmental moisture plays a critical role in the intensification of tropical cyclones and hurricanes (both classified as forms of tropical storms) by facilitating convection and precipitation outside the storm system. In a study on the impact of environmental moisture on tropical cyclone intensification, Longtao Wu, of the California Institute of Technology, illustrated this concept, stating:

“Enhanced environmental moisture ahead of a northwestward-moving storm induces a dry air intrusion into the inner core, limiting storm intensification. In contrast, increased moisture in the rear quadrants favors intensification by supplying additional moisture to the inner core and promoting storm symmetry, with primary contributions originating from moisture increases in the boundary layer. The varying impacts of environmental moisture on storm intensification are governed by the relative locations of moisture perturbations and their interactions with the storm's Lagrangian structure”. [28]

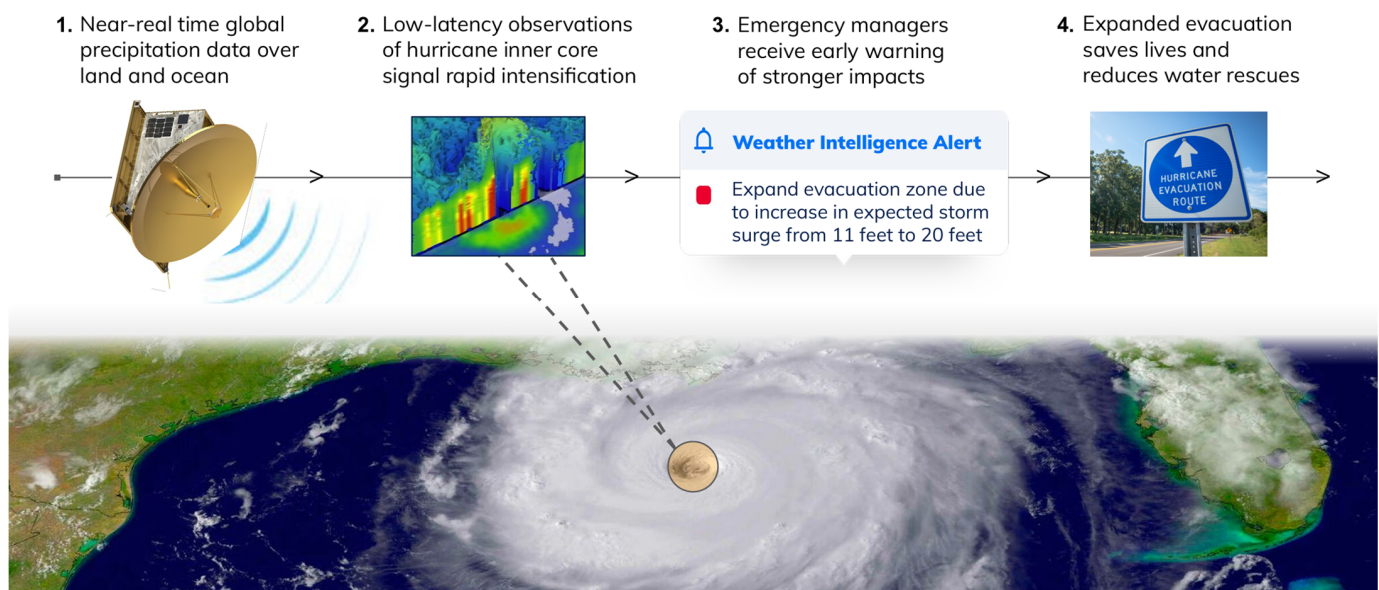


Figure 6. Weather Intelligence Alert Flowchart—Courtesy of Tomorrow.io.

This demonstrates that a precipitation model continuously updated with near-real-time data driven by high satellite revisit rates can significantly improve the ability to predict severe weather events and issue timely warnings to decision-makers. Continuous tracking of atmospheric moisture properties will enhance forecasting accuracy—one of many applications that, over time, will become a practical daily reality thanks to initiatives like Tomorrow.io’s satellite constellation.

5.2. A Global Measurements Integration Model

Tomorrow.io’s technology enables the integration of all acquired measurements into a unified, cloud-centric global model. This approach ensures that weather forecasting models and their outputs are accessible worldwide, facilitating the development of numerous new solutions and use cases, as outlined below:

1. Data Acquisition

Measurements are continuously collected worldwide by Tomorrow.io’s multi-sensor satellite constellation, which utilizes various remote sensing technologies onboard.

2. Data Transmission to Ground Stations

The acquired data is relayed to a distributed network of commercial ground stations, such as those operated by Kongsberg Satellite Services (KSAT). KSAT, a Norwegian commercial entity, manages a global network of ground stations, specializing in receiving downlinks from polar-orbiting satellites and providing Earth observation services [29]. Its infrastructure includes 25 physical sites strategically located to ensure comprehensive coverage for polar-orbiting satellites.

3. Cloud-Based Data Processing and Accessibility

The data received from the ground station network is transferred to a commercial cloud infrastructure, such as Amazon Web Services (AWS), as seen in Figure 7. This allows raw data to be processed by cloud-based analytics tools and ensures that weather forecasting models are accessible to any organization worldwide, enabling seamless integration into various services and applications.

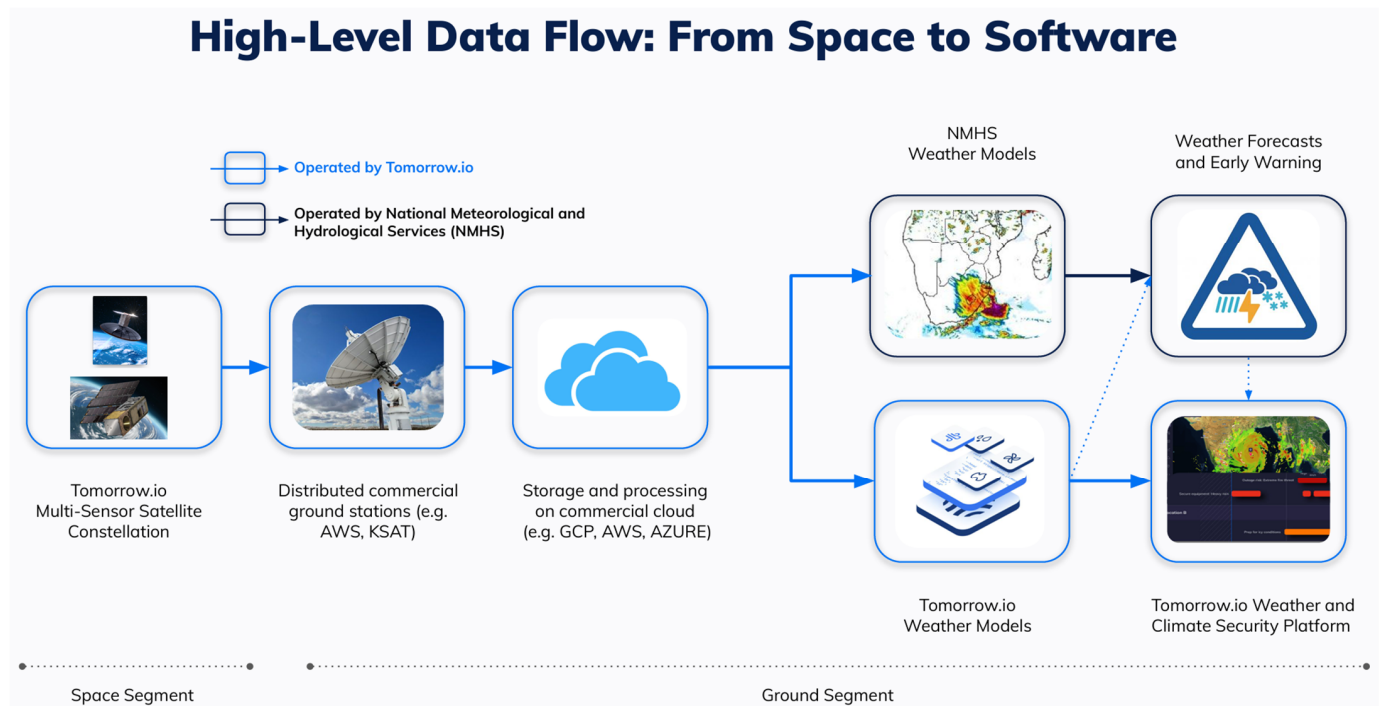


Figure 7. From Space to Software—Flowchart—Courtesy of Tomorrow.io.

4. Data Processing Using Advanced Forecasting Models

In the pre-final stage, vast amounts of collected data are fed into advanced weather forecasting models that utilize Artificial Neural Networks (ANNs) specifically designed [30] for remote sensing applications (see Figure 7). These models are required to operate in near-real time to enable the timely production of severe weather event warnings and other mission-critical outputs.

Among the techniques likely employed are statistical regression approaches, which use large datasets of input-output pairs to empirically derive statistical relationships between atmospheric parameters. In simple cases, linear regression techniques based on second-order statistical moments (such as covariances) can be used to compute a linear fit that minimizes the sum of squared errors between the model and the observed data. However, due to the inherently complex and nonlinear nature of atmospheric systems, linear models are often insufficient. Consequently, nonlinear regression methods, including ANNs, are better suited for capturing the complex, high-dimensional relationships present in remote sensing datasets [31]. ANNs, as a specialized class of nonlinear regression operators, have proven highly effective for remote-sensing-based weather forecasting applications [32].

5. Dissemination to End-Users

Finally, model predictions are integrated into Tomorrow.io's weather and climate security platform, where they are made accessible to end-users. This enables organizations to receive actionable, location-specific weather forecasts and severe weather alerts, enhancing preparedness and decision-making capabilities across industries.

One could argue that such a model represents a significant advancement beyond the current capabilities of the Advanced Weather Interactive Processing System (AWIPS2), developed as an extension of the RSA Project under the National Oceanic and Atmospheric Administration (NOAA). AWIPS2 primarily serves to visualize site-specific weather data, providing decision-makers and monitoring personnel with improved situational awareness [33].

The emergence of near-real-time weather forecasting technologies is expected to have profound implications, not only for civilian sectors but also for defense and military operations.

For example, military missions involving Remotely Piloted Aircrafts (RPAs) have experienced substantial losses over the past three decades, amounting to over \$100 million in taxpayer expenses. Given that RPAs are unmanned, they rely heavily on accurate environmental awareness to operate safely and effectively. A significant portion of these losses can be attributed to limitations in conventional weather forecasting technologies, which fail to provide the required resolution and timeliness for dynamic mission environments.

Therefore, the integration of near-real-time weather forecasting capabilities into military operations could serve as a critical support tool for RPA operators, significantly reducing operational losses and enhancing mission safety. Moreover, by delivering high-resolution, accurate weather data remotely, this technology may reduce or even eliminate the need for deploying dedicated weather sensors at or near mission locations [34].

6. Profound Impact Potential on Decision-Making Processes

Weather and environmental monitoring have always been critical to informing strategic decision-making in both civilian and military domains. With legacy systems such as the Defense Meteorological Satellite Program (DMSP) nearing the end of their operational lifespans, there is growing urgency to modernize satellite constellations and remote sensing capabilities. This section synthesizes major advancements across several programs, notably the Geostationary Operational Environmental Satellite-R (GOES-R) series and the Weather System Follow-on Microwave (WSF-M) satellite program.

6.1. Modernizing Military Weather Monitoring

The U.S. military depends heavily on space-based meteorological systems to support Joint All-Domain Command and Control (JADC2) operations, which require near-real-time, high-resolution meteorological data to enable agile, informed responses. Existing limitations in terrestrial sensors, combined with the vast and complex Indo-Pacific theater, underscore the need for advanced space-based weather solutions [35].

The Weather System Follow-on Microwave (WSF-M) program directly addresses capability gaps left by the aging DMSP constellation. The first WSF-M satellite was launched aboard a SpaceX Falcon 9 rocket in April 2024. This satellite enhances the military's ability to measure ocean surface winds, monitor tropical cyclone development, and detect charged atmospheric particles. Beyond military applications, WSF-M provides valuable environmental data, including sea ice mapping, soil moisture content, and snow depth measurements. Future plans include launching a second WSF-M satellite in 2028, complemented by additional platforms developed under the Electro-Optical/Infrared Weather System (EWS) program, scheduled for deployment in 2025 and 2027 [36]. These programs ensure redundancy, resilience, and continuity in space-based weather monitoring [37].

6.2. Enhancing National Resilience with the GOES-R Series

The Geostationary Operational Environmental Satellite-R (GOES-R) series, developed collaboratively by NOAA, NASA, and Lockheed Martin, represents a significant leap in geostationary weather monitoring and forecasting. Since the launch of GOES-R in 2016, the series has dramatically enhanced the precision, frequency, and scope of atmospheric observations, providing critical data to improve disaster preparedness, severe weather warnings, and long-term climate assessments [38].

Notable technological advances include:

- (a) Geostationary Lightning Mapper (GLM): Enables continuous real-time lightning detection and tracking.

- (b) Solar Ultraviolet Imager (SUVI): Facilitates monitoring of solar activity and space weather conditions.
- (c) Compact Coronagraph (CCOR): Provides early detection of coronal mass ejections, helping protect satellites, power grids, and communication systems.

The GOES-R series offers up to four times greater temporal and spatial resolution compared to previous generations, delivering near-instantaneous updates crucial for timely responses to severe weather events [39]. The final satellite in the series, GOES-U, was launched aboard a SpaceX Falcon Heavy rocket in June 2024 from NASA's Kennedy Space Center. GOES-U extends advanced weather and environmental monitoring capabilities across the eastern hemisphere, spanning from the west coast of Africa to New Zealand, solidifying the series' role in supporting national resilience and global environmental intelligence [40].

6.3. Leveraging AI for Climate Research

IBM's collaboration with NASA demonstrates the growing role of artificial intelligence in advancing climate research and Earth system science. With a partnership legacy dating back to 1949, supporting programs such as Apollo and various Earth science missions, IBM has played an instrumental role in space exploration and data analysis. In 2022, IBM and NASA formalized their collaboration under a Space Act Agreement to address global climate challenges. A major outcome of this partnership is the development of the HLS Geospatial Foundation Model (HLS Geospatial FM), trained on NASA's extensive geospatial datasets, including Harmonized Landsat and Sentinel-2 imagery. Leveraging IBM's Vela supercomputer infrastructure, this model significantly enhances the efficiency of geospatial data analysis, enabling critical applications such as flood prediction, land-use monitoring, and disaster response [41].

Key technical advancements include:

- (a) A 15% improvement in accuracy compared to traditional models.
- (b) A fourfold increase in geospatial analysis speed, achieved with reduced reliance on labeled training data.
- (c) Broad applicability to tasks such as agricultural impact analysis, deforestation monitoring, and wildfire tracking.

In alignment with open science principles, IBM has made the HLS Geospatial FM available on Hugging Face, fostering global collaboration and accelerating innovation in Earth observation.

6.4. Advances in Solar Weather Forecasting

Solar weather events, including solar flares and coronal mass ejections, pose significant risks to satellite operations, space-based communication networks, and space domain awareness. Accurate solar weather forecasting is essential, as solar events can alter atmospheric density, leading to increased drag on satellites—particularly those in very low Earth orbit (vLEO)—and can disrupt critical operations such as tracking adversarial space assets during geomagnetic storms. Booz Allen Hamilton's researchers have developed advanced solar weather forecasting models by leveraging data from NASA's Solar Dynamics Observatory (SDO), which provides continuous high-resolution solar imagery. Using autoencoder-based machine learning algorithms, these models efficiently process vast solar datasets, detecting patterns indicative of emerging solar activity. Trained on over 300,000 high-resolution solar images spanning more than a decade, these models have improved the prediction accuracy of solar phenomena, including sunspots and flares [42].

By adopting an open-source development framework, researchers ensure continuous model refinement through the integration of diverse datasets. These AI-driven models

contribute to enhanced forecasting of atmospheric drag effects on satellites, more precise orbit determination, and unified frameworks for space domain awareness.

6.5. China's Advancements in Meteorological Satellite Constellations

China has made significant strides in disaster monitoring and weather forecasting through its Fengyun (FY) satellite series and the complementary Tianmu-1 and Yunyao-1 [43] constellations. The FY-4 series, in particular, has reduced typhoon recognition times from 15 min to just 5 min and has substantially improved track prediction accuracy.

These advancements have strengthened global numerical weather prediction (NWP) capabilities, enhanced severe weather monitoring, and improved the detection of environmental changes and ecological hazards. Furthermore, China's meteorological satellites actively support international collaboration, providing data to over 115 countries, reinforcing China's position as a leader in meteorological innovation and ecological sustainability [44].

7. Conclusions

Over the past 60 years, remarkable progress has been achieved in the field of remote sensing and its applications for weather and climate forecasting, primarily driven by advancements in satellite technologies. Despite these accomplishments, the future of this domain holds even greater potential—constrained only by our imagination—as technological breakthroughs continue to redefine the limits of what is possible.

The deployment of near-real-time weather forecasting models is expected to have a transformative impact across numerous sectors. In the defense sector, where weather conditions critically influence operational success, such capabilities are of strategic importance for mission planning and execution across air, land, and maritime domains [45]. Similarly, in civilian sectors, including emergency response and disaster management, accurate and timely weather forecasts have proven essential, enabling earlier warnings and more effective evacuation procedures that can save lives [37].

These advancements have been made possible by the collective efforts of international institutions and private entities, including the World Meteorological Organization (WMO), national meteorological administrations, NOAA [46], Tomorrow.io, aerospace corporations [47], IBM, Booz Allen Hamilton, and key contributors from China and Europe. Their innovations—ranging from improved satellite constellations to AI-powered forecasting models—represent critical milestones in our ongoing pursuit of enhanced environmental awareness and disaster resilience. Ultimately, the driving force behind these technological achievements remains clear: to safeguard human life by improving our ability to predict, monitor, and respond to increasingly complex weather and climate phenomena.

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